**25901212** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

## Fourth SEMESTER

## Discipline Specific Core Course (DSC) - MG4DSCECO201 - QUANTITATIVE ECONOMICS - II

(2024 ADMISSION ONWARDS)

**Duration: 2 hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which of the following is not true about statistics? [U] / [CO1]

   a). Aggregate facts b). Numerical facts

   c). Affected by many causes d). Individual information

2. The median, deciles, percentiles are all considered as _____ [A] / [CO1]

   a). Mathematical average b). Systematic average

   c). Positional average d). Representative average

3. Which concept links PMF and PDF conceptually? [Ap] / [CO2]

   a). Both measure randomness b). Both represent probability distributions

   c). Both require integration d). Both apply only to integers

4. CPI measures changes in: [K] / [CO1, CO4]

a). Wholesale prices b). Export prices

c). Retail prices paid by consumers d). Industrial prices

5. The values of F-distribution are always _____. [U] / [CO1]

a). Negative b). Zero only

c). Positive d). Both positive and negative

6. Semi-average method divides data into [Ap] / [CO2]

a). Two parts b). Three parts

c). Four parts d). Random parts

7. In a scatter diagram, which variable is typically plotted on the horizontal (x) axis? [U] / [CO1, CO4]

a). The dependent variable b). The independent variable

c). The variable with the larger values d). The variable measured second

8. When should the median be preferred over the mean? [U] / [CO1]

a). In data with outliers b). In symmetric data

c). In small data sets d). In normal distribution

9. Multiplicative model is used when variations are [Ap] / [CO2]

a). Constant b). Proportional

c). Negative d). Irregular

10. A study shows a correlation of r = 0.90 between ice-cream sales and drowning incidents. What can be concluded: [U] / [CO1, CO4]

a). Eating ice-cream causes drowning

b). Drowning causes an increase in ice-cream sales

c). There is a strong positive correlation, but it does not imply causation

d). The correlation is invalid because ‘r’ greater than 0.5

**[1x10 = 10]**

## Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. What is the difference between a simple bar diagram and a multiple bar diagram? [Ap] / [CO2]

12. Write the Z-score formula and explain its use. [A] / [CO3]

13. The largest value is 80 and the smallest is 20. Find the Coefficient of Range. [Ap] / [CO2]

14. Two teachers rank 6 students. Teacher X gives ranks: 1,2,3,4,5,6 and Teacher Y gives ranks: 1,3,2,5,4,6. Analyse whether the teachers mostly agree or disagree. [An] / [CO1, CO4]

15. Define multistage sampling [Ap] / [CO2]

16. Write the formula for binomial distribution [Ap] / [CO2]

17. Find the Mean Deviation from Mean for: 7, 9, 11, 13. [Ap] / [CO2]

18. If r = -0.85 between price and demand, interpret this result. [A] / [CO1, CO4]

19. Distinguish between a symmetrical and a skewed distribution with a suitable explanation. [U] / [CO1]

20. Give two economic examples of time series (GDP, stock price, etc.) to explain moving average. [Ap] / [CO2]

21. Define finite population. [Ap] / [CO2]

22. Find the standard deviation, coefficient of variation, and variance for the number of books read by a student in a month. [Ap] / [CO2]

| Books | 3 | 6 | 9 | 12 | 15 | 18 | 21 | 24 | 27 |
|---|---|---|---|---|---|---|---|---|---|
| No. of students | 7 | 10 | 8 | 6 | 4 | 7 | 12 | 5 | 9 |

23. Give few examples of economic time series. [Ap] / [CO2]

**[3x10 = 30]**

## Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

24. What is secondary data? Elaborate the sources of secondary data along with its merits and demerits. [Ap] / [CO2]

25. The income of 8 families is: 20, 25, 30, 35, 40, 45, 50, 100. Find (a) Range (b) Quartile Deviation [Ap] / [CO2]

26. Establish the meaning of Probability sampling and explain the major probability sampling methods. [Ap] / [CO2]

27. Calculate median from the following: [K] / [CO1]

| Marks (less than) | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 |
|---|---|---|---|---|---|---|---|---|
| No. of Students | 3 | 11 | 16 | 32 | 58 | 70 | 76 | 79 |

28. Write a detailed note on properties of the normal distribution curve. [Ap] / [CO2]

29. From the given data, calculate Index number using Fisher's formula and Marshall- Edgeworth formula [A] / [CO1, CO4]

| Commodity | Price in 2005 | Quantity in 2005 | Price in 2010 | Quantity in 2010 |
|---|---|---|---|---|
| A | 3 | 10 | 5 | 12 |
| B | 6 | 5 | 7 | 4 |
| C | 8 | 4 | 9 | 6 |

30. From the following data; [A] / [CO3]

1. Write down the two regression equations.
2. Obtain the best estimate of y when, x = 60 and
3. Obtain the best estimate of x when, y = 70.

$\bar{x}$ = 65, $\bar{y}$ = 67, $\sigma x$ = 2.5, $\sigma y$ = 3.5, r = 0.8

**[6x5 = 30]**